

## Modbus communication for Polytropic heat pumps:

### IVS / IVN / IVP / IVR made in 2019 and after.

#### Important:

For all machine produced in 2022 and after, the Modbus is used for Polyconnect gateway (Wi-Fi and/or 4G).

In order to use the Modbus, gateway have to be disconnected.

The consequence of this operation is that there will be no Polyconnect service anymore on this machine:

- Polyconnect APP will not be available anymore.
- Polyconnect Pro APP will not be available anymore.
- All additional options linked with Polyconnect will not be available anymore, such as:
  - o Eventual warranty extension
  - o Proactive after sales
  - o All future options depending on Polyconnect service.

#### External Gateway

Polytropic provide information to control the heat pump through Modbus RS485 RTU with a compliant gateway. Polytropic do not provide this gateway and so cannot be held responsible for any software or its programming.

By using this connection installer/integrator take the responsibility of the gateway configuration and working.

#### Modbus communication information

Modbus / RS485 Remote Terminal Unit half-duplex

- Baud rate: 9600 baud
- Start bit: 1
- Data bit: 8
- Parity check: None
- End bit: 1

Connection need ground: 3 terminals on the PCB:

- A (Data +)
- B (Data -)
- GND (ground)

The PCB of the heat pump is always slave.

Default slave address is 17.

The Modbus slave address of the heat pump can be changed using address #47 (between 17 and 32).

#### Additional information:

Addresses are base10, protocol is base 0 and machine needs first MSB.

## Modbus address table:

**Note: do never try to read/write data on any other address of the bus, irreparable damage may be caused to the unit. Repairs to this type of damage will not be covered by the warranty!**

| Address |     | Description                                                                  | Byte info    |             | Range                                                                                                                                                                                                                        | Error message / display |
|---------|-----|------------------------------------------------------------------------------|--------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 47      | R/W | PCB Modbus slave address                                                     |              | Double byte | 17 - 32                                                                                                                                                                                                                      | -                       |
| 57      | R/W | Water inlet compensation temp                                                |              | Double byte | -90 – 90 (-9°C - 9°C)                                                                                                                                                                                                        | -                       |
| 58      | R/W | Maximum value available for water temp. setting                              |              | Double byte | 25 - 60 (°C)                                                                                                                                                                                                                 | -                       |
| 59      | R/W | Optional circulation pump mode                                               |              | Double byte | Range: 0~2<br>0: ON=ON, Stand-by = OFF<br>1: Compressor OFF = water pump OFF after 2 minutes delay; water pump ON 30 second before compressor ON<br>2: Water pump ON 5 min every 60 minutes, if heat pump ON: water pump ON. | -                       |
| 500     | R   | High pressure switch status                                                  | word0_Bit00  | Double byte | 0 - 1                                                                                                                                                                                                                        | HP                      |
|         |     | Low pressure switch status                                                   | word0_Bit01  |             | 0 - 1                                                                                                                                                                                                                        | LP                      |
|         |     | Reserved                                                                     | word 0_Bit02 |             | 0 - 1                                                                                                                                                                                                                        | -                       |
|         |     | Water flow switch status                                                     | word0_Bit03  |             | 0 - 1                                                                                                                                                                                                                        | Icon "water flow"       |
|         |     | Reserved                                                                     | word0_Bit04  |             | 0 - 1                                                                                                                                                                                                                        | -                       |
|         |     | Optional water pump status                                                   | word0_Bit05  |             | 0 - 1                                                                                                                                                                                                                        | Icon "water pump"       |
|         |     | Optional electric heater status                                              | word0_Bit06  |             | 0 - 1                                                                                                                                                                                                                        | -                       |
|         |     | Four-way valve status                                                        | word0_Bit07  |             | 0 - 1                                                                                                                                                                                                                        | -                       |
|         |     | Bottom plate heater status                                                   | word0_Bit08  |             | 0 - 1                                                                                                                                                                                                                        | -                       |
|         |     | Compressor heater status                                                     | word0_Bit09  |             | 0 - 1                                                                                                                                                                                                                        | -                       |
| 501     | R   | Water inlet temp. sensor failure                                             | word1_Bit0   | Double byte | 0 - 1                                                                                                                                                                                                                        | AL03                    |
|         |     | Water outlet temp. sensor failure                                            | word1_Bit1   |             | 0 - 1                                                                                                                                                                                                                        | AL04                    |
|         |     | Coil temp sensor failure                                                     | word1_Bit2   |             | 0 - 1                                                                                                                                                                                                                        | AL05                    |
|         |     | Ambient temp. sensor failure                                                 | word1_Bit3   |             | 0 - 1                                                                                                                                                                                                                        | AL06                    |
|         |     | Ambient temp. too low protection                                             | word1_Bit4   |             | 0 - 1                                                                                                                                                                                                                        | AL14                    |
|         |     | Compressor outlet temp sensor failure                                        | word1_Bit5   |             | 0 - 1                                                                                                                                                                                                                        | AL01                    |
|         |     | Compressor inlet temp sensor failure                                         | word1_Bit6   |             | 0 - 1                                                                                                                                                                                                                        | AL02                    |
|         |     | Temp difference between inlet and outlet too high protection                 | word1_Bit7   |             | 0 - 1                                                                                                                                                                                                                        | AL15                    |
|         |     | Temp difference between inlet and outlet too high protection (3 times in 1h) | word1_Bit8   |             | 0 - 1                                                                                                                                                                                                                        | AL16                    |
|         |     | Cooling overcool protection                                                  | word1_Bit9   |             | 0 - 1                                                                                                                                                                                                                        | AL17                    |
|         |     | High pressure protection                                                     | word1_Bit10  |             | 0 - 1                                                                                                                                                                                                                        | AL10                    |
|         |     | High pressure protection (3 times in 1h)                                     | word1_Bit11  |             | 0 - 1                                                                                                                                                                                                                        | AL11                    |
|         |     | Low pressure protection                                                      | word1_Bit12  |             | 0 - 1                                                                                                                                                                                                                        | AL12                    |
|         |     | Low pressure protection (3 times in 1h)                                      | word1_Bit13  |             | 0 - 1                                                                                                                                                                                                                        | AL13                    |
|         |     | Water flow switch protection                                                 | word1_Bit14  |             | 0 - 1                                                                                                                                                                                                                        | FLO                     |
|         |     | Anti-freezing protection                                                     | word1_Bit15  |             | 0 - 1                                                                                                                                                                                                                        | -                       |

| Address |     | Description                              | Byte info        |             | Range                      | Error message / display                                                                                                                                                     |
|---------|-----|------------------------------------------|------------------|-------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 502     | R   | Compressor out temp. too high protection | word2_Bit0       | Double byte | 0 - 1                      | AL18                                                                                                                                                                        |
|         |     | AC voltage input protection              | word2_Bit1       |             | 0 - 1                      | AL19                                                                                                                                                                        |
|         |     | DC main line current protection          | word2_Bit2       |             | 0 - 1                      | AL21                                                                                                                                                                        |
|         |     | AC current input protection              | word2_Bit3       |             | 0 - 1                      | AL20                                                                                                                                                                        |
|         |     | Compressor over-current protection       | word2_Bit4       |             | 0 - 1                      | AL22                                                                                                                                                                        |
|         |     | IPM temp. overheat protection            | word2_Bit5       |             | 0 - 1                      | AL23                                                                                                                                                                        |
|         |     | IPM protection                           | word2_Bit6       |             | 0 - 1                      | AL24                                                                                                                                                                        |
|         |     | compressor driver protection             | word2_Bit7       |             | 0 - 1                      | AL25                                                                                                                                                                        |
|         |     | DC fan motor 1 failure                   | word2_Bit8       |             | 0 - 1                      | AL09                                                                                                                                                                        |
|         |     | DC fan motor 2 failure                   | word2_Bit9       |             | 0 - 1                      | -                                                                                                                                                                           |
|         |     | Driver PCB communication failure         | word2_Bit10      |             | 0 - 1                      | EA07                                                                                                                                                                        |
| 503     | R   | Main PCB communication failure           | word3_Bit00      | Double byte | 0 - 1                      | AL07                                                                                                                                                                        |
|         |     | EE data read error                       | word3_Bit01      |             | 0 - 1                      | AL08                                                                                                                                                                        |
|         |     | Reserved                                 | word3_Bit2-Bit14 |             | null                       | -                                                                                                                                                                           |
|         |     | Defrost                                  | word3_Bit15      |             | 0 - 1                      | Icon "Defrost"                                                                                                                                                              |
| 510     | R   | Sensor: Compressor out temp              | word10           | Double byte | -300 - 2200 (-30 – 220 °C) | Value                                                                                                                                                                       |
| 511     | R   | Sensor: Compressor In temp               | word11           | Double byte | -300 - 2200 (-30 – 220 °C) | Value                                                                                                                                                                       |
| 512     | R   | Sensor: Water in temp                    | word12           | Double byte | -300 - 2200 (-30 – 220 °C) | Value                                                                                                                                                                       |
| 513     | R   | Sensor: Water out temp                   | word13           | Double byte | -300 - 2200 (-30 – 220 °C) | Value                                                                                                                                                                       |
| 514     | R   | Sensor: Coil temp                        | word14           | Double byte | -300 - 2200 (-30 – 220 °C) | Value                                                                                                                                                                       |
| 515     | R   | Sensor: Ambient temp                     | word15           | Double byte | -300 - 2200 (-30 – 220 °C) | Value                                                                                                                                                                       |
| 516     | R   | Sensor: IPM temp                         | word16           | Double byte | -300 - 2200 (-30 – 220 °C) | Value                                                                                                                                                                       |
| 521     | R   | Power supply: voltage (AC)               | word21           | Double byte | 0 - 500 (VAC)              | -                                                                                                                                                                           |
| 522     | R   | Power supply: current (AC)               | word22           | Double byte | 0 - 100 (A)                | -                                                                                                                                                                           |
| 523     | R   | Compressor failure code                  | word23           | Double byte | 0 - 65535                  | EA01-EA19                                                                                                                                                                   |
| 1000    | R/W | Operation mode                           | word0_Bit0-Bit3  | Double byte | 0-7                        | 0 = Standby<br>1 = Auto (heat & cool)<br>2 = Cooling + Smart<br>3 = Cooling + Boost<br>4 = Cooling + Eco<br>5 = Heating + Smart<br>6 = Heating + Boost<br>7 = Heating + Eco |
|         |     | Unit ON/OFF                              | word0_Bit4       |             | 0 - 1                      | "Stand-by" icon                                                                                                                                                             |
|         |     | Reserved                                 | word0_Bit5-Bit15 |             | null                       | -                                                                                                                                                                           |
| 1001    | R/W | Water temp setting                       | word1            | Double byte | 0 – 600 (0 – 60°C)         | Value                                                                                                                                                                       |